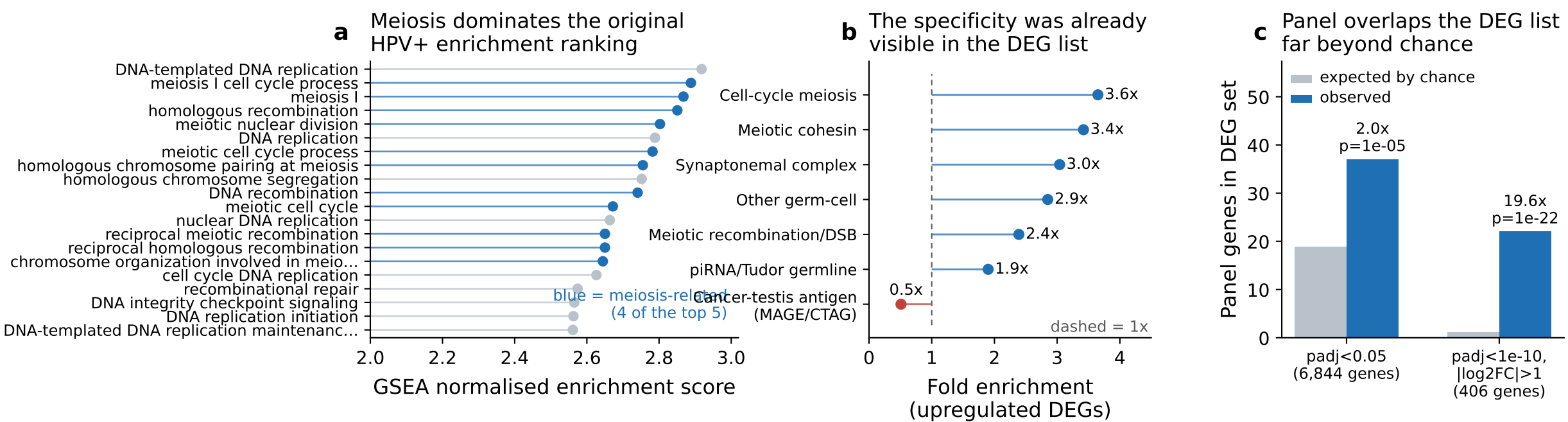
a
Meiosis dominates the original HPV+ enrichment ranking
DNA-templated DNA replication
meiosis I cell cycle process
meiosis I
homologous recombination
meiotic nuclear division
DNA replication
meiotic cell cycle process
homologous chromosome pairing at meiosis
homologous chromosome segregation
DNA recombination
meiotic cell cycle
nuclear DNA replication
reciprocal meiotic recombination
reciprocal homologous recombination
chromosome organization involved in meio...
cell cycle DNA replication
recombinational repair
DNA integrity checkpoint signaling
DNA replication initiation
DNA-templated DNA replication maintenanc...
blue = meiosis-related
(4 of the top 5)
2.0
2.2
2.4
2.6
2.8
3.0
GSEA normalised enrichment score
b
The specificity was already visible in the DEG list
Cell-cycle meiosis
Meiotic cohesin
Synaptonemal complex
Other germ-cell
Meiotic recombination/DSB
piRNA/Tudor germline
Cancer-testis antigen (MAGE/CTAG)
3.6x
3.4x
3.0x
2.9x
2.4x
1.9x
0.5x
dashed = 1x
0
1
2
3
4
Fold enrichment (upregulated DEGs)
c
Panel overlaps the DEG list far beyond chance
expected by chance
observed
2.0x
p=1e-05
19.6x
p=1e-22
Panel genes in DEG set
0
10
20
30
40
50
padj<0.05
(6,844 genes)
padj<1e-10,
|log2FC|>1
(406 genes)